<u>**Visualizing Key Predictors of MLB Team Win Rate**</u>

Team #100: Heath Cameron, Andrew Zimmerman, Samuel Solberg, Patrick Crowe, Ian Crane

1. Introduction

In baseball and more specifically Major League Baseball, the ability to accurately predict outcomes is imperative in the statistics heavy and driven sporting event. The problem that Team 100 has committed to solving is developing a model and visualization for key predictors that are pivotal in determining the win rate of an MLB team. This outcome is important because total wins in a season is a key focus for stakeholders in the sport. Total wins are a direct measure of the overall success of a team. More wins increase the likelihood of making the postseason and ultimately playing in and winning the World Series. Front offices and analysts care about understanding which factors truly drive wins and how roster construction translates into results. Fans and media care about reliable win projections and clear explanations of why teams succeed. Bettors care about identifying mispriced win totals and uncovering statistical edges before sportsbooks adjust. Most prior work focuses on predicting individual game outcomes rather than full season wins. Even strong models achieve only about 60% accuracy at the game level, highlighting how difficult baseball is to predict (Soto Valero, 2016). Many studies also rely on limited time periods, older data, or a narrow set of features, which may not reflect modern MLB dynamics (Richards & Guell, 1998; Barry & Hartigan, 1993). Instead of predicting games one at a time, we will directly model season-level win totals.

2. Problem Definition

As it currently stands, there is a need to understand how salary and team performance metrics impact total season wins, while also being able to perform exploratory data analysis on those metrics by tuning the parameters. It is well understood that paying a large sum of money for good players tends to lead to more wins and having players that can score lots of runs tends to increase total team wins, but are there are minute factors that may not be being considered? Understanding the threshold for how much money should be spent to build a team and what performance factors should be considered will lead to increased insight on total team wins.

3. Literature Survey

Season-level outcome prediction in Major League Baseball (MLB) has been studied extensively using a wide range of statistical models. Early choice models used aggregate team level statistics to predict a discrete outcome, such as a team winning their division (Barry & Hartigan, 1993). Bill James' Pythagorean Formula was developed to estimate season win totals based on run differentials, showing a correlation between a team's performance and season win totals (Miller, 2005). Given the MLB's lack of salary cap and the perceived pay to win dynamic, earlier studies focused on using regression to explore relationships between payroll and win percentage, season attendance levels, and the likelihood of winning a division or a World Series (Richards & Guell, 1998; Yilmaz & Chatterjee, 2003). A limitation of earlier models is their reliance on offensive attributes and lack of defensive or pitching attributes (Richards & Guell, 1998; Yilmaz & Chatterjee, 2003). Recent research has begun to close this gap, using Markov regression or SVM models with team offensive, defensive, and pitching variables of prior seasons as inputs to predict the winning percentage of the upcoming season (Choi & Ji, 2025; Soto Valero, 2016).These models utilized team season level totals in lieu of more granular player specific attributes and focused more on showing explanatory relationships (i.e. salary, in-season improvements, etc.) rather than predicting the win/loss record for a team's entire schedule.

Studies comparing methods like SVM to traditional linear regression models have shown a slight improvement in the accuracy of predicting the outcome of an MLB game by 3-4% (Allen & Savala, 2025; Soto Valero, 2016). The increasing popularity of using statistical and machine learning models for predicting outcomes of MLB games has also led to an increase in the number of stats, or features that are collected for each player and team. For example, the Statcast system is a publicly available dataset showing features for every pitch including bat speed, angle of attack, and other biometric data (Bailey et al., 2020). As a result of this increase in available data, researchers are applying feature selection algorithms such as recursive feature elimination to enhance the interpretability of the model, visualizations, and prediction accuracy (Pandey & Gupta, 2025; Li et al., 2022; Soto Valero, 2016). While some models forecast full season win totals, most focus on other predictions such as season batting average predictions, and predicting run-line bet winners.

Wins Above Replacement (WAR) is one of the most common representations of a player's contribution to a team. WAR represents the added wins a player provides to their team when compared to a replacement (i.e., a free agent or someone off the bench) and is calculated using individual player statistics for a given season (Robinson, 2014). Historically, WAR favors offense, and several studies have been conducted to show and correct some of the historical positional imbalances by adding more positional context to the WAR calculation, specifically for players who excel at defense or at pitching as opposed to offense (Brill & Wyner, 2024; Ehrlich et al., 2021). Other research investigates the impact of often excluded in-game factors such as the number of stolen base attempts, the timeliness of in-game substitutions, and relief pitcher matchups on team performance (Demmink, 2010; Hirotsu & Wright, 2005). However, these player valuations approaches, or individual player attributes, are rarely used in models to forecast the season win totals.

4. Proposed Method

Our team innovated on the existing literature and methodologies in a few ways. First, in the literature we reviewed, we noted that most studies focused their evaluation solely on either some category of player/team performance metrics or financial metrics. In our approach, we sought to evaluate if and how these groups of predictors interact with one another to explain wins. This ultimately led to the innovation of two side by side, separate, but complementary models to predict total season wins with one being salary metrics and the other being performance metrics. This, in addition to the interactivity included, gives users a clear visual and tangible demonstration of our model results. In particular, it highlights the very low explanatory power of salary alone in comparison to player metrics. This challenges the viewpoint that the lack of salary constraints in the MLB allows the largest teams to buy success. Additionally, our interactive dashboard enables easy exploratory analysis, giving enhanced insight into the relevant factors surrounding maximization of total team wins in major league baseball and help stakeholders better predict outcomes.

Our group will build our own predictive regression model to estimate how many games a given MLB team will win in a season. In addition to generating predictions, we will develop an interactive visualization that allows users to explore which team performance, roster, and financial factors are most strongly associated with winning. Instead of predicting individual games, we will directly model season-level win totals. Our approach will combine team performance metrics, payroll structure, and efficiency measures that prior research has shown to relate to success (Choi & Ji, 2025; Richards & Guell, 1998). By leveraging contemporary MLB data, our model improves earlier studies like Barry and Hartigan (1993), whose findings were based on decades-old datasets. By pairing the model with a visualization layer, we also improve transparency and interpretability.

Initially, the data needed to be developed for analyzing. We leveraged the Lahman Baseball Database (Lahman, 2025) that is contained on the Society for American Baseball Research website. They offer an SQL version, Microsoft Access version, and a comma-delimited version. Our team used the comma-delimited version, as it is simple to import and apply basic Engineer-Transform-Load (ETL) operations to the data. The data set itself is quite comprehensive, containing essentially all relevant statistics in relation to Major League Baseball from the years 1871 to 2025. We decided to focus on the years 2010 to 2016, since the approach has a focus on payroll structure and its impact on total wins, which 2016 is the last year that salaries have been updated for this dataset. Next, we aggregated the data by team and rolled up all the relevant statistics to that level. It is important to note that all salary values were normalized to prevent impact from year-over-year changes like inflation. This step left us with the following features that became the key focus for the linear regression analysis.

Table 1 – Salary and Performance Model Features and Weights

Salary Model Feature	**Feature Weight**
intercept	79.573285
salary_skew	-2.71531150658
salary_std	-0.00000058333
reliever_pitcher_payroll	0.00000001451
starter_pitcher_payroll	0.00000009869
batting_payrolll	0.00000007514
Performance Model Feature	**Feature Weight**
intercept	45.85181
all_star_count	0.469335031209
attendance	0.000000673843
ERA	-17.729029402642
HR	0.106290814450
team_batting_avg	363.413715741908
E	-0.039436001546

From the tables, there are two developed models that can be used to predict total wins. These complementary models can be used to predict wins for a team. The first, Salary Model, leverages financial features for the teams to provide insight into how payroll can impact performance. The second, Performance Model, indicates which team performance metrics need to be focused on and developed, providing insight into how a team should spend money to achieve those statistics. The R^2 score was used to evaluate the accuracy of the models, and those results can be seen below. The current R^2 benchmark for predictive models in Major League Baseball on average is approximately 0.65 and the current model outperforms this indicating that we have made an improvement in the predicative capability for total win models in Major League Baseball.

Finally, the visualization will be built on top of the dataset and the model weights. We will allow the features to act as sliders in a dashboard and give the user the power to toggle the visual and show the potential performance of their team in relation to all other teams given the two models. A basic visual of this dashboard can be seen in figure 1 and figure 2. We employed Shiny for Python to deploy our interactive dashboard, and it provides the functionality needed to display two models showing predicted vs. actual wins for the team with respect to both models, the win distribution of the data that is embedded in the application, relative payroll vs. wins, salary skew vs. wins, and access to the dataset itself. By providing an interactive visualization for key stakeholders, this will help to drive adoption by the end-user, leading to better decision making. This visualization will continue to be optimized for enhancing user experience and user interface based on beta testing that will be completed over the next few weeks.

Figure 1 – MLB Win Predictor Visualization on Shiny for Python – Salary Model

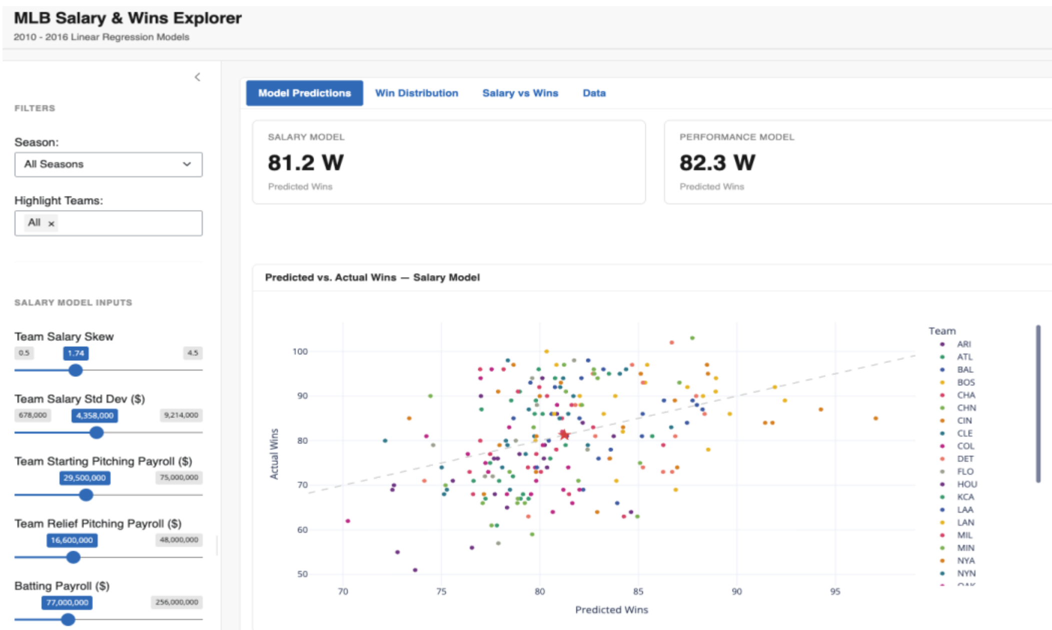
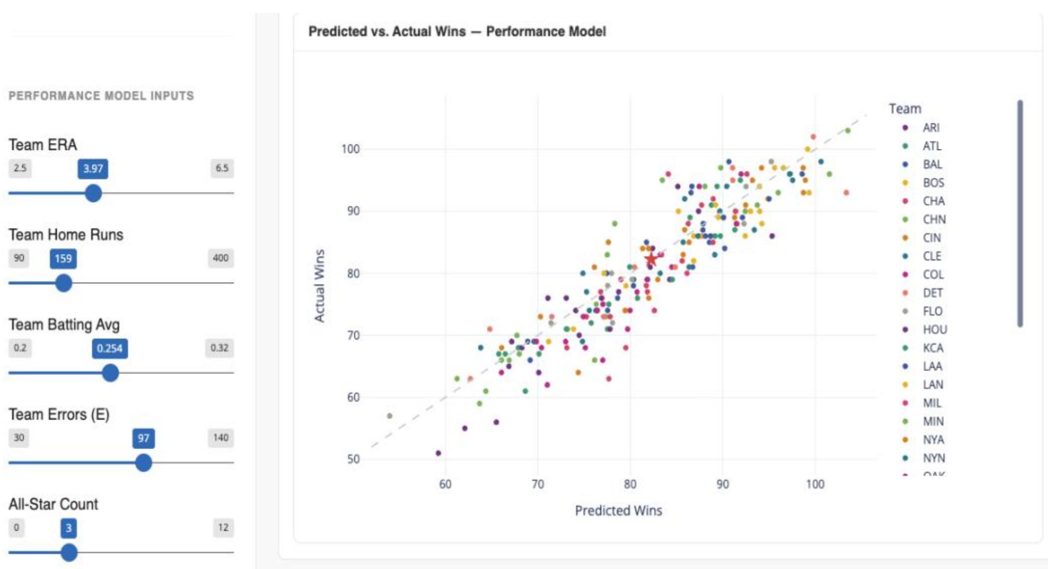


Figure 2 – MLB Win Predictor Visualization on Shiny for Python – Performance Model



5. Evaluation

As noted in our literature review, we identified an R^2 of 0.65 as the high-end benchmark for similar MLB predictive models. We used this number to evaluate the improvement over existing models introduced by our approach. Our evaluation involved assessing a range of models with various variable selection techniques and benchmarking these models against this 0.65 R^2. For the testbed for our experiments, we extracted and transformed the Lahman Baseball Database tables for pitching, hitting, manager, fielding, and salary statistics into a data frame in R (Lahman, 2025). For 2010-2016 data we generated several models' using manual variable selection (Using VIF), backwards elimination, and forwards selection and compared resulting model statistics. Our experiments sought to answer the following questions:

- Which statistics contribute the most to the explanatory power of our win-predicting model?
- Will using payroll statistics contribute significantly to the model's predictive power when combined with player performance?
- Can salary alone be used to effectively predict MLB team success?
- Does including a mix of offensive, fielding, and pitching statistics lead to a stronger predictive model?

Several of our experimental models achieved an R^2 greater than 0.65. In general, our models that blended payroll and player performance statistics had an R^2 ranging from 0.58 to 0.86. Within this group, 5 models produced an R^2 between 0.82 and 0.86. These highest performing models all included ERA (pitching metric), Errors (fielding metric), and some combination of batting and salary predictors. Each of these models handled salary differently. One model used total salary, while others used transformed salary metrics, like salary standard deviation, the ratio of pitcher to batter salaries, and the skew of salary towards the top handful of players on a team. Perhaps surprisingly, while all five of these models included some form of salary even when using backwards elimination or forwards variable selection, we found that the salary metrics included had little meaningful impact on the predictive power of the models as described by R^2. In fact, the highest performing experimental model, which had an R^2 of 0.86, exhibited an R^2 of 0.85 with salary removed as a predictor. This demonstrated to us that using payroll statistics did not contribute significantly to the model's predictive power when combined with player performance, but including a mix of offensive, fielding, and pitching statistics did lead to a stronger predictive model than relying on offense alone.

This turned our attention to the question of salary's explanatory power over wins. We hypothesized that the lack of predictive contribution to our models was due to multicollinearity, as higher salaries would be paid to players with better statistics. This led us to create a series of experiments using the same testbed but only using financial metrics as predictors. We discovered that salary alone had very low predictive power in comparison to other metrics. Our best model at this stage produced an R^2 of just 0.15 using a variety of payroll metrics (See Table 1). Our conclusions from these models' experiments led to visualization of the two models separately. This gives users a visual representation of the difference in impact between the two types of predictors used between the models.

Table 2 – Model R^2 Results

Model	R^2
Salary Model	0.15
Performance Model	0.85
Industry Standard Model	0.65

6. Conclusions and Discussion

In summary, creating two separate linear regression models related to salary metrics and performance metrics, respectively, was determined to be a unique and successful methodology for predicting total season wins. The model outperforms benchmark industry models and focuses on only the most relevant factor's for predicting total season wins. The most unique element of this project is the visualization built on top of the linear regression model. The dashboard allows the user to toggle the relevant features for a hypothetical team and evaluate how those factors impact the team's performance in relation to total season wins. Having this capability, the user can fine tune the parameters and see exactly where events like breakeven or diminishing returns occur. These are powerful insights for not only a league manager or front office employee, but also a passionate fan or the sports betting fanatic.

It is important to also consider the limitations that our model and visualization will face when deploying it in the real world. Our model focused on data that was generated during the 2010 to 2016 MLB baseball season. This reason was due to the lack of more recent data, and it is not understood why this data is lacking from the dataset. However, future work should focus on updating the model regarding more recent data to determine if there is a change over time, due to external factors not included in the model, that impacts the predictive capability of the model. Other limitations within the current model are the lack of flexibility in adding other features to the dashboard. In a future iteration of the project, it may be worthwhile to make all available features in major league baseball at the disposal of the person interacting with the model so that they may pick and choose what attributes contribute to the total season wins, while learning more about how these features may impact total season wins.

All team members have contributed a similar amount of effort. Throughout the project, team members committed to different elements of the final goal. One week a team member would be looking for data sources or researching different project topics, the next week they would be writing a portion of the report, and another week they would be coding up regression models. We found that working together to communicate about the different project needs and getting exposure to the different tasks was a valuable experience that helped to further improve us as professionals.

References

Allen, M., & Savala, P. (2025). *Assessing win strength in MLB win prediction models*. ArXiv.org. https://arxiv.org/abs/2511.02815

Bailey, S. R., Loeppky, J., & Swartz, T. B. (2020). The Prediction of Batting Averages in Major League Baseball. *Stats*, *3*(2), 84–93. https://doi.org/10.3390/stats3020008

Barry, D., & Hartigan, J. A. (1993). Choice Models for Predicting Divisional Winners in Major League Baseball. *Journal of the American Statistical Association*, *88*(423), 766–766. https://doi.org/10.2307/2290761

Brill, R. S., & Wyner, A. J. (2024). Introducing Grid WAR: rethinking WAR for starting pitchers. Journal of Quantitative Analysis in Sports, 20(4), 293–329. https://doi.org/10.1515/jqas-2023-0095

Choi, S. H., & Ji, S.-K. (2025). A Study of Winning Percentage in the MLB Using Fuzzy Markov Regression. *Mathematics*, *13*(6), 1008. https://doi.org/10.3390/math13061008

Demmink, H., III. (2010). Value of stealing bases in Major League Baseball: “Stealing” runs and wins. Public Choice, 142(3/4), 497–505.

D. Pandey and R. Gupta, “VisMLB: An Hybrid Data Visualization Model for Major League Baseball Matches Using Tableau,” Proceedings of the 2025 3rd IEEE International Conference on Industrial Electronics: Developments & Applications (ICIDeA), 2025, https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10962958

Ehrlich, J., Perline, J., Potter, J., & Sanders, S. (2021). Does a salary premium exist for offensive output in Major League Baseball? *Managerial Finance*, *47*(3), 326–335. https://doi.org/10.1108/MF-04-2020-0186

Hirotsu, N., & Wright, M. (2005). Modelling a baseball game to optimise pitcher substitution strategies incorporating handedness of players. IMA Journal of Management Mathematics, 16(2), 179–194. https://doi.org/10.1093/imaman/dpi009

Li, S.-F., Huang, M.-L., & Li, Y.-Z. (2022). Exploring and Selecting Features to Predict the Next Outcomes of MLB Games. *Entropy*, *24*(2), 288. https://doi.org/10.3390/e24020288

Miller, S. J. (2005). *A Derivation of the Pythagorean Won-Loss Formula in Baseball*. https://doi.org/10.48550/arxiv.math/0509698

Richards, D. G., & Guell, R. C. (1998). Baseball success and the structure of salaries. *Applied Economics Letters*, *5*(5), 291–296. https://doi.org/10.1080/758524403

Robinson, S. J. (2014). Extracting individual offensive production from baseball run distributions. International Journal of Computer Science in Sport, 13(2), 20–33. https://iacss.org/wp-content/uploads/2024/09/IJCSS-Volume13_2014_Edition2.pdf

Soto Valero, C. (2016). Predicting Win-Loss outcomes in MLB regular season games – A comparative study using data mining methods. *International Journal of Computer Science in Sport*, *15*(2), 91–112. https://doi.org/10.1515/ijcss-2016-0007

Yilmaz, M. R., & Chatterjee, S. (2003). Salaries, Performance, And Owners' Goals In Major League Baseball: A View Through Data. *Journal of Managerial Issues*, *15*(2), 243–255. JSTOR. https://doi.org/10.2307/40604428

Lahman, S. (2025). *Lahman Baseball Database* (Version 2025) [Data set].